

Name: M.Sai Chaitanya

ID number: 2520090023

Week 10

1. Investigate inode structures using commands such as `ls -li`, `stat`, and `find`. Create hard links and symbolic links and explain their impact on inode allocation.

```
saichaitanya@GeethzzSai: ~/f × + ▾
saichaitanya@GeethzzSai:~$ cd Practical_week10
saichaitanya@GeethzzSai:~/Practical_week10$ pwd
/home/saichaitanya/Practical_week10
saichaitanya@GeethzzSai:~/Practical_week10$ echo "Hello from Chaitanya" > original_chaitanya.txt
saichaitanya@GeethzzSai:~/Practical_week10$ cat original_chaitanya.txt
Hello from Chaitanya
saichaitanya@GeethzzSai:~/Practical_week10$ ls -li original_chaitanya.txt
51837 original_chaitanya.txt
saichaitanya@GeethzzSai:~/Practical_week10$ stat original_chaitanya.txt
  File: original_chaitanya.txt
  size: 21          Blocks: 8          IO Block: 4096   regular file
Device: 8,48      Inode: 51837         Links: 1
Access: (0644/-rw-r--r--)  Uid: ( 1000/saichaitanya)   Gid: ( 1000/saichaitanya)
Access: 2026-09-26 14:49:08.292429937 +0000
Modify: 2026-09-26 14:48:59.484433287 +0000
Change: 2026-09-26 14:48:59.484433287 +0000
 Birth: 2026-09-26 14:48:59.484433287 +0000
saichaitanya@GeethzzSai:~/Practical_week10$ Inode: XXXXX
Links: 1
Inode:: command not found
Links:: command not found
saichaitanya@GeethzzSai:~/Practical_week10$ ln original_chaitanya.txt hardlink_chaitanya.txt
saichaitanya@GeethzzSai:~/Practical_week10$ ls -li original_chaitanya.txt hardlink_chaitanya.txt
51837 -rw-r--r-- 2 saichaitanya saichaitanya 21 Sep 26 14:48 hardlink_chaitanya.txt
51837 -rw-r--r-- 2 saichaitanya saichaitanya 21 Sep 26 14:48 original_chaitanya.txt
saichaitanya@GeethzzSai:~/Practical_week10$ stat original_chaitanya.txt
  File: original_chaitanya.txt
  size: 21          Blocks: 8          IO Block: 4096   regular file
Device: 8,48      Inode: 51837         Links: 2
Access: (0644/-rw-r--r--)  Uid: ( 1000/saichaitanya)   Gid: ( 1000/saichaitanya)
Access: 2026-09-26 14:49:08.292429937 +0000
Modify: 2026-09-26 14:48:59.484433287 +0000
Change: 2026-09-26 14:50:25.080399457 +0000
 Birth: 2026-09-26 14:48:59.484433287 +0000
```

```

saichaitanya@GeethzzSai:~/Practical_week10$ stat original_chaitanya.txt
  File: original_chaitanya.txt
  size: 46          Blocks: 8          IO Block: 4096   regular file
Device: 8,48      Inode: 51837         Links: 2
Access: (0644/-rw-r--r--)  Uid: ( 1000/saichaitanya)   Gid: ( 1000/saichaitanya)
Access: 2026-09-26 14:51:15.000379333 +0000
Modify: 2026-09-26 14:51:08.860381534 +0000
Change: 2026-09-26 14:51:08.860381534 +0000
 Birth: 2026-09-26 14:48:59.484433287 +0000
saichaitanya@GeethzzSai:~/Practical_week10$ ln -s original_chaitanya.txt sym
link_chaitanya.txt
saichaitanya@GeethzzSai:~/Practical_week10$ ls -li original_chaitanya.txt ha
rdlink_chaitanya.txt symlink_chaitanya.txt
51837 -rw-r--r-- 2 saichaitanya saichaitanya 46 Sep 26 14:51 hardlink_chaita
nya.txt
51837 -rw-r--r-- 2 saichaitanya saichaitanya 46 Sep 26 14:51 original_chaita
nya.txt
52796 lrwxrwxrwx 1 saichaitanya saichaitanya 22 Sep 26 14:51 symlink_chaitan
ya.txt -> original_chaitanya.txt
saichaitanya@GeethzzSai:~/Practical_week10$ ls -i original_chaitanya.txt
51837 original_chaitanya.txt
saichaitanya@GeethzzSai:~/Practical_week10$ find . -inum 13732
saichaitanya@GeethzzSai:~/Practical_week10$ find . -inum 51837
./hardlink_chaitanya.txt
./original_chaitanya.txt
saichaitanya@GeethzzSai:~/Practical_week10$
saichaitanya@GeethzzSai:~/Practical_week10$
saichaitanya@GeethzzSai:~/Practical_week10$
saichaitanya@GeethzzSai:~/Practical_week10$ rm original_chaitanya.txt
saichaitanya@GeethzzSai:~/Practical_week10$ cat hardlink_chaitanya.txt
Hello from Chaitanya
Modified using hard link
saichaitanya@GeethzzSai:~/Practical_week10$ cat symlink_chaitanya.txt
cat: symlink_chaitanya.txt: No such file or directory
saichaitanya@GeethzzSai:~/Practical_week10$

```

2.Design a program using mmap() to perform file reading and writing. Compare memory-mapped 1/0 with traditional read/write operations in terms of efficiency and Implementation complexity.

```
saichaitanya@GeethzzSai: ~/f × + v
GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>
#include <sys/mman.h>
#include <sys/stat.h>
#include <string.h>

int main() {
    int fd;
    struct stat st;
    char *data;

    fd = open("mmap_input.txt", O_RDWR);

    if (fd < 0) {
        perror("open");
        return 1;
    }

    if (fstat(fd, &st) < 0) {
        perror("fstat");
        close(fd);
        return 1;
    }

    data = mmap(NULL, st.st_size,
                PROT_READ | PROT_WRITE,
                MAP_SHARED, fd, 0);

    if (data == MAP_FAILED) {
        perror("mmap");
        close(fd);
        return 1;
    }

    printf("Original file content:\n%s\n", data);

    strcpy(data, "Modified using mmap!");

    printf("File modified successfully.\n");

    msync(data, st.st_size, MS_SYNC);

    munmap(data, st.st_size);
    close(fd);

    return 0;
}
```

Output:

```
saichaitanya@GeethzzSai:~/Practical_week10$ echo "Hello from mmap! This file will be modified." > mmap_input.txt
saichaitanya@GeethzzSai:~/Practical_week10$ cat mmap_input.txt
Hello from mmap! This file will be modified.
saichaitanya@GeethzzSai:~/Practical_week10$ nano src/mmap_file.c

saichaitanya@GeethzzSai:~/Practical_week10$ gcc src/mmap_file.c -o bin/mmap_file
saichaitanya@GeethzzSai:~/Practical_week10$ ./bin/mmap_file
Original file content:
Hello from mmap! This file will be modified.

File modified successfully.
```